

Orb v1

Clean tyre-degradation curves from Friday practice, scored against the race every Sunday.

Part A: the Challenge Day roadmap, hour by hour, with deliverables, done-criteria, dependencies, fallbacks and the rubric criterion each task serves. Part B: the technical manual, simplified: what we are building, what it does, how it works, what we chose and why, and twenty questions a mentor or jury will ask about technical depth.

Team Orb v1 · Samuel Christ · TrackShift 2026, Tyre Degradation Intelligence · Plaksha University, 12 to 13 September 2026

Numbers in this document come from the locked results file generated 2026-09-12T21:33:10 IST: 11 scored 2026 weekends, 29 compound-weekends, every figure leave-one-weekend-out. The estimator and the first six weekends are disclosed pre-work from the idea round (4 to 5 September); everything else was built on Challenge Day.

Contents

- **Part A. Roadmap:** A1 where we stand; A2 the two hours before building; A3 Saturday build schedule; A4 Sunday; A5 cut list and decision rules; A6 rubric map; A7 after the event
- **Part B. Technical manual:** 1 summary; 2 the problem; 3 inputs and outputs; 4 how it works; 5 results; 6 choices with pros and cons; 6b competitive landscape with eight questions; 8 twenty jury questions with answers; appendix

Part A. Roadmap

Clock times are IST on Saturday 12 September unless marked Sunday. Build start T0 = 11:15. Anchors that cannot move: Madrid FP3 16:00 to 17:00 (refresh_fp3.log not found), qualifying 19:30 to 20:30 (data landed 21:33), race Sunday 18:30, after the event closes. Windows below are the plan; the Status column records what landed, read from the refresh logs and the published forecast. Priorities: P0 must exist for the pitch; P1 makes the pitch stronger; P2 is cut first. Rubric letters: T technical depth 30%, F problem-solution fit 25%, D demo 20%, S scalability and real-world viability 15%, C communication 10%.

A1. Where we stand at 09:15, updated with what landed since

Component	Status	Evidence
Data: 2026 rounds	Done: all 14 rounds that exist in the timing data, incl. qualifying sessions; Bahrain and Saudi Arabia do not exist for 2026	feat/ (70 session files); China FP1 position feed degraded and refused
Data: 2025 rounds	Partial: 10 of 14 complete; Belgium (was mislabelled, deleted), Britain, Hungary, Zandvoort, Monza pending on FastF1's 500-calls-per-hour limit; automatic retry every 25 min from 09:28	feat2025/, retry2025.log
Pipeline and lock	Done in the morning; lock rebuilt after FP3 (time not on record) and after qualifying (21:33) with 11 scored weekends; generated 2026-09-12T21:33:10	pipeline.py, out/lock.json, refresh_fp3.log, refresh_q.log
Estimator, gate, fallback, factors, bands	Done and validated leave-one-weekend-out	out/validation.csv, sensitivity.json
Push diagnostic and second opinion	Done	lock: push_diagnostic, path_b
Strategy layer with compound offsets from practice or qualifying medians	Done; replay scored on 10 weekends	strategy2.py, lock: strategy
Liquid model with fair ablation	Done (v1 and tuned v2)	out/liquid.json, liquid_v2.json
Dashboard, six views, URL-addressable	Done; last verified 07:40; must be re-launched on the venue laptop	app.py, launch.json
Deck v1 (10 slides), talk track, figures	Done in the morning; figures and the 10-slide deck rebuilt by refresh.sh at each refresh (last 21:33); talk track regenerated from the 21:33 lock by this build	out/Orb_v1_ChallengeDay.pdf, talk_track.md, fig_*.png
Madrid live forecast	Issued from FP1 and FP2 in the morning; qualifying refresh landed 21:33 with FP1, FP2, FP3 and Q in the lock; hashed forecast published 21:33 IST (sha256 8e5489d5824a)	lock: live.Madrid; refresh_fp3.log, refresh_q.log; out/forecast_Madrid_2026.json and .sha256
This manual	Done	out/Orb_v1_Roadmap_and_Manual.pdf

A2. The two hours before building (09:15 to 11:15): research, planning and environment only

When	Task	Done when
09:15 to 09:45	Read Part B once. Mark anything you cannot explain in one sentence; ask me before 11:15.	A list of at most five open questions
09:45 to 10:15	Environment on the venue network: open the dashboard (launch config clearstint-dashboard, port 8501), load ?ev=Madrid&view=Weekend and ?view=Season%20validation, confirm both render. Check retry2025.log after 10:00.	Dashboard renders both views; 2025 retry status known
10:45 to 11:15	Memorise the 90-second mentor version and the eight prepared answers (talk_track.md). Write the one-sentence disclosure: estimator and six-weekend validation were pre-work disclosed in the idea round; everything else was built here. Confirm the presentation slot and check-in with the organisers.	Script said aloud twice under 90 s

A3. Saturday build schedule from T0

ID	Window	Min	Prio	Task	Deliverable	Done when	Depends	Fallback	Status
A1	11:15 to 11:40	25	P0 T F	Rebuild the lock with every 2026 weekend that now has a race (adds Monaco; China stays refused). Record every number that changed against deck v1.	python pipeline.py; diff summary printed; a changes.md with old and new numbers	Dashboard shows the new lock time; changes.md exists	None	If a new weekend breaks a rule, exclude it and say why in changes.md	
A2	11:40 to 12:10	30	P1 T	Band coverage check: share and count of 90% bands containing the race value, leave-one-out, issued and fallback separately. Add to the lock and to the validation view.	validation.band_coverage in lock; one line on the validation view	Numbers appear on the dashboard	A1	If coverage is poor, widen the factor resampling and report both	
A3	12:10 to 12:40	30	P1 D	Static HTML export of the dashboard: one self-contained file per view with the Plotly figures and tables, for the case the app cannot run on the venue network.	out/static/*.html	Opens from the file system with no server	A1	Skip if the venue network and the app are stable at 12:00	
A4	12:30 to 13:00	30	P0 F	If a detailed problem brief is released: map every requirement to a view, a number, or a gap. Re-order the rest of this table if needed.	requirements.md with a three-column map	Every requirement has an owner row	Brief	If no brief is released, use the public three-sentence statement	
B1	13:00 to 13:45	45	P1 T C	Barcelona miss: why the soft prediction ran low and the naive line implied the right plan. One paragraph and one figure. Honest misses are a slide.	notes/barcelona.md, fig_barcelona.png	Paragraph reads in under 30 s	A1	Cut to one sentence on the strategy slide	
B2	13:45 to 14:15	30	P1 T	Data-quality slide: the lap-by-lap feed check, Hungary race and China FP1 refused, exclusion counts per reason per weekend.	fig_quality.png	Figure is legible at slide size	A1	Use the exclusions view live instead	
C1	14:15 to 15:15	60	P0 D C	Mentor checkpoint package: the demo path as five URLs in order, the 90-second script with A1 numbers, the eight prepared answers, the disclosure sentence. Rehearse once at the desk with the dashboard.	demo_path.md; script v2	Desk run under 90 s with no dead clicks	A1	None; this is the checkpoint	
C2	15:15 to 16:00	45	P0 C	Deck v2: update every number from the new lock; add the data-quality slide; keep ten slides.	out/Orb_v1_ChallengeDay_v2.pdf	Every number matches the lock	A1 B2	Keep v1 and correct numbers verbally	

ID	Window	Min	Prio	Task	Deliverable	Done when	Depends	Fallback	Status
D1	16:00 to 17:20	80	P1 T S	While FP3 runs: 2025 same-track prior table if the 2025 data completed (does the medium factor repeat at the same circuits?); practice-to-sprint test on the six sprint weekends (does Friday predict the Saturday sprint too?).	out/prior2025.csv, out/sprint_test.csv, one figure	Two tables with leave-on e-out numbers	2025 data	If 2025 data is still short, run the sprint test only	
D2	17:30 to 18:00	30	P0 D F	Madrid FP3 refresh: refresh.sh, re-issue the curves, record what moved and why (expect the medium ramp to shrink as drivers settle; hard may appear). Update the live view and the deck slide.	lock with FP3; madrid_changes.md	Live view shows FP3 in sessions_in_data	FP3 data	If data is late, refresh at 18:30; the FP1+FP2 issue stands	
E2	18:45 to 19:30	45	P2 T	Liquid tab final: tuned v2 numbers, cliff-lap rule sanity check on three weekends, one sentence per weekend on shape.	lock or liquid.json updated; tab text	Tab numbers match liquid_v2.json	None	First to cut	
F1	19:30 to 20:45	75	P0 S C	While qualifying runs: README with the disclosure paragraph, method summary, commands, data sources and limits; make the repository public; requirements.txt pinned.	Public repo URL	Fresh clone runs pipeline.py on the feature CSVs	None	Private repo shared with judges on request	
F2	20:45 to 21:30	45	P0 D F	Madrid qualifying refresh: compound offsets re-derived from practice or qualifying medians (nominal step where implausible), final strategy call (plan, stint lengths, crossover, what flips it), publish the forecast as JSON and a one-page PDF with a timestamp and a SHA-256 hash, and post the hash somewhere public.	out/forecast_Madrid.json and .pdf; hash posted	Hash matches the file; timestamp before 21:45	Qualifying data, D2	If qualifying data is late, publish on FP3 numbers and say so	Landed 21:33: FP1, FP2, FP3, Q in the lock; forecast issued 21:33 IST, sha256 8e5489d5824a
G1	21:30 to 22:30	60	P0 C	Deck v3 with the final Madrid slide and the forecast hash; talk track v3.	Orb_v1_ChallengeDay_v3.pdf	Numbers match the lock and the forecast	F2	v2 plus a verbal update	Talk track regenerated from the 21:33 lock by this build
G2	22:30 to 23:15	45	P0 C	One full timed five-minute run with the dashboard, then fix every dead click and every sentence over 20 words.	Run under 5:00	Recorded time	G1	None	
G3	23:15 to 06:00	405	P0	Sleep. A solo presenter with four hours of sleep loses more points than any P1 task gains.	Rest	You slept	None	None	

A4. Sunday 13 September

ID	Window	Prio	Task	Done when
H1	06:00 to 07:00	P0	Deck final check against the lock; print two copies of the deck and one of Part B section 8.	No number differs
H2	07:00 to 08:00	P0	Two timed runs; second one with a friend or another team asking the eight questions.	Both under 5:00
H3	08:00 to 08:30	P0	Dashboard smoke test on the venue network; static export open in a second tab.	Five demo URLs load
H4	08:30 to slot	P1	Buffer. If the 2025 prior finished overnight, one sentence on the scalability slide. Nothing new is built.	Nothing broken
H5	Presentation	P0	Five minutes, then Q&A from Part B section 8. Open the dashboard on Madrid live, end on the forecast hash.	Done
I1	After 18:30 race	P1	Score Madrid with the same estimator; publish the scorecard whatever it says, with the pre-race hash beside it.	Scorecard public

A5. Cut list and decision rules

- Cut in this order if behind: E2 liquid tab, D1 sprint and 2025 tests, A3 static export (only if the app is stable), B1 written note (keep one sentence). Never cut A1, C1, C2, D2, F2, G1, G2, G3, H2, H3.
- If the venue network fails: the pipeline and dashboard run offline from the cached data; only FP3 and qualifying refreshes need the network, and they can wait for a phone hotspot.
- If FastF1 rate-limits again: the two Madrid refreshes are a handful of calls each and take priority over any 2025 download; kill retry2025.sh before a refresh.
- If the 12:30 brief asks for something not on this table: it goes in at A4 and displaces the lowest-priority task still ahead of it.
- If a number changes after A1: the deck, the talk track and the manual are regenerated from the lock, never edited by hand.

A6. Rubric map: which tasks carry which criterion

Criterion	Weight	Carried by	What the jury sees
Technical depth and innovation	30%	Validation ladder, fifth confounder, ablation, coverage (A1, A2, B1, D1, E2)	Leave-one-weekend-out numbers; a measured confounder that, in the public methods and rival repositories we reviewed, we did not find stripped elsewhere; an honest network test
Problem-solution fit	25%	Curves from practice, gate and fallback, post-race scorecard, Madrid live (A1, D2, F2)	The brief answered literally: clean Friday curves and a validation tool that scores them on Sunday
Demo and working prototype	20%	Dashboard, forecast publication (C1, F2, H3)	Six views running on real data, a live weekend
Scalability and real-world viability	15%	Any Grand Prix by name from free data; F2, F3, F1 Academy, Indian F4; beyond the track (F1, D1)	One command per weekend; a customer with two engineers and public timing
Presentation and communication	10%	Deck v3, talk track, rehearsals (C2, G1, G2, H2)	Five minutes, one story: Friday lies twice

A7. After the event

- Week 1: score Madrid and publish; package as a command-line tool (clearstint <Grand Prix>) that produces the lock and the static views for any weekend since 2018.
- Weeks 2 to 4: band coverage over a full season; hierarchical transfer factor with partial pooling across 2025 and 2026; per-lap track temperature; sprint races as a second validation target; OpenF1 intervals for races with degraded position feeds.

- Month 2: timing adapters for F2, F3 and F1 Academy; a pilot with a junior team or a broadcaster; beyond the track, the same clean-then-transfer loop on a public battery ageing set, then a fleet pilot in Punjab.

Part B. Technical manual

1. One-page summary

The problem. Teams want to know how fast each tyre compound will degrade on Sunday, and Friday practice is the only place to measure it before the race. But a Friday lap time is not a tyre measurement. It carries fuel burning off, the track getting faster as rubber goes down, traffic, the car and driver, and how hard the driver was pushing at that moment. A straight line through Friday laps at the 2026 Hungarian Grand Prix says the medium degrades at +0.285 s per lap of tyre age. The race showed +0.042. Seven times wrong.

What Orb v1 does. It takes the free public timing and telemetry for any Grand Prix weekend and runs six steps: **ingest** every lap of every session; **clean** each lap by removing fuel, track evolution, traffic and the stint it belongs to; **gate** so that a curve is issued only when Friday really supports one; **learn**, from previous weekends only, how much Sunday shrinks the Friday number for each compound; **forecast** the race degradation curve with a band, the crossover laps and a one-stop versus two-stop call; and **score** itself against the race with the same estimator on Sunday night.

The proof. Across 11 weekends and 29 compound-weekends, with every number computed leave-one-weekend-out, the naive line has a mean error of 0.137 s/lap against the race. Orb v1 has 0.023. Correlation with the race goes from 0.20 to 0.78. It beats the naive fit on 28 of 29 cases. 16 curves were issued; 13 compounds were withheld and forecast as low degradation instead; the withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13, and the fallback error was 0.023 s/lap against 0.106 for the naive line over the 13 withheld cases.

The discovery. We measure the energy a driver puts through the tyre from the public 3.7 Hz traces: where the cleaned Friday slope pointed the wrong way (9 cases), the driver was ramping up in every one; 4 of the 13 withheld compounds show a falling energy trend and were withheld for too few clean laps. The median within-run energy trend is +0.10 MJ per lap of age on withheld compounds and 0.00 on issued ones. That is a fifth confounder, the driver's push profile; in the public methods and rival repositories we reviewed, we did not find one that strips it. It is measured, not assumed.

Live. Madrid's curves were issued this morning from FP1 and FP2 alone, refreshed after FP3 and after qualifying (landed 21:33, FP1, FP2, FP3 and Q in the lock) to the values shown: soft +0.068 s/lap, issued, 90% band -0.11 to +0.26, 46 clean laps; medium +0.028 s/lap, withheld, 90% band -0.01 to +0.09, 112 clean laps (withheld because drivers were learning a new circuit; low degradation forecast by two independent routes); hard: no curve, 4 clean laps on Friday (excluded-laps table). Hashed forecast published 21:33 IST (sha256 8e5489d5824a); it will be scored after Sunday's race.

The product. One locked results file feeds a six-view dashboard, the strategy replay, and the deck. It runs for any Grand Prix by name from free data on a laptop.

2. The problem in plain words

Friday lies twice. The first lie is about what the tyre did on Friday. Every lap time mixes five things:

- **Fuel.** A 2026 car burns roughly 1.1 kg per lap, and each kilogram is worth about 0.030 s. A long run gets faster by about 0.033 s every lap for that reason alone, which looks like negative degradation.
- **Track evolution.** A green track gets faster as rubber goes down, by 0.6 to 0.8 s per hour at Madrid on Friday (FP1 and FP2 in the lock). Laps later in the session look better than they are.
- **Traffic.** Running within 60 m of another car costs downforce and lap time, and it is not the tyre's fault.
- **The stint.** Different drivers, cars, fuel loads and engine modes produce different absolute lap times. Only the change within a stint is about the tyre.
- **The push profile.** Drivers ramp up through a long run when learning a track or nursing a hard compound, and ease off when a soft tyre is going away. The lap time follows the driver as much as the tyre.

The second lie is about Sunday. Even a perfectly cleaned Friday curve is not Sunday's curve, because drivers attack tyres in practice and manage them in a race. We measured this with the same estimator on both sides of the weekend: across the season the medium's race degradation runs at about 0.45 times its cleaned Friday value, the soft transfers roughly one to one ($\times 0.97$), and the hard factor (median $\times 1.17$) is not applied because weekends disagree. This ratio is learned from previous weekends and applied only where weekends agree.

Why it matters. A degradation curve is a decision: when to pit, which compound, one stop or two. Replayed on Hungary with the race-derived curves as the reference, the plan implied by the naive Friday line loses 74 s against the best plan; Orb v1's plan loses 24 s. Replayed on the 10 scored 2026 weekends (3 of them sprint weekends), the naive plan costs between 0 and 155 s, Orb v1's between 0 and 55 s, and Orb v1 beats the naive plan on 8 of the 10 scored weekends; the two misses are Barcelona (+55 s against +0 s for naive) and Australia (+25 s against +14 s for naive).

3. What the product does: inputs and outputs

Inputs (all free and public):

- FastF1 (v3.8): every lap of every session with compound, tyre age, stint, track status, timing accuracy flags, deleted-lap flags from race control, pit in and out, weather once a minute, and car telemetry (speed, throttle, brake, gear, DRS) plus position at about 3.7 Hz, including the distance to the car ahead.
- OpenF1 (optional): gap-to-car-ahead every four seconds in races, pit-lane durations, stints, race-control messages. Used for cross-checks and for races whose position feed is degraded.
- Pirelli compound nominations; practice and qualifying laps for compound pace offsets.

Outputs, per weekend and per compound:

- **Issued curve:** race degradation in s per lap of tyre age, a 90% band, the naive and cleaned Friday slopes for comparison, the transfer factor and how many weekends it was learned from, and the basis in one sentence.
- **Withheld:** the gate reason (too few clean laps, or no positive degradation signal in cleaned practice), a low-degradation forecast with a band, the push diagnostic (energy trend through the runs), and a push-adjusted second opinion where available.
- **Exclusions:** every practice lap the estimator dropped, with its reason, and the counts per reason.
- **Strategy:** compound offsets from practice or qualifying medians (nominal Pirelli-range step where the measured step is implausible), crossover laps, the best one-stop and two-stop plans with stint lengths, and, once the race is run, the time cost of following each Friday view instead of the best plan.
- **Season validation:** the calibration ladder (naive, cleaned, Orb v1), per-compound errors, bootstrap intervals, the list of withheld cases and what the race did, the sensitivity sweep.

Dashboard views: Weekend (live or scorecard), Strategy, Liquid model, Season validation, Excluded laps, Method & assumptions. Each view is addressable by URL for the demo. Static HTML fallback planned in case the

app cannot run.

4. How it works, step by step

4.1 Data and quality checks

For every lap we compute: lap and sector times, tyre age and compound, session time, track status, whether the lap was deleted, pit in/out flags, a traffic share (fraction of the lap spent within 60 m of the car ahead, from the distance-to-car-ahead channel), a tyre-energy proxy, and two feed-quality numbers: how many distinct position points the lap contained and what share of speed samples were stale repeats. Feeds that are degraded at source, such as the Hungary race (about 26 distinct position points per lap instead of about 300) and China FP1, are refused for telemetry-based quantities rather than modelled silently.

4.2 The cleaning model

Laps are first filtered: unknown or wet compounds, deleted laps, pit laps, laps under yellow, safety car or red flags, laps flagged inaccurate by timing, laps without telemetry, laps with more than 30% of the distance in traffic, runs shorter than five clean laps, and laps slower than 105% of the run's best (cool-down and aborted laps) are dropped. Every drop is recorded with its reason.

Then one regression, in words: *lap time = a constant for the stint + degradation rate for the compound × tyre age + a fuel term + a track-evolution term*. The stint constant absorbs the driver, the car, the fuel load and the engine mode, so only the change within a stint informs the degradation rate. The fuel term is a stated prior from the 2026 regulations (1.1 kg per lap at 0.030 s per kg), not estimated from practice, because within a stint fuel and tyre age move together and cannot be separated; a public estimate from race data gives 0.0294 s per kg. Track evolution is measured per session from every driver's push laps (laps within 1% of that driver's own best) against session time, and subtracted. The same regression, with fuel from race progress instead of the prior, is applied to the race. Because the estimator is identical on both sides, the practice-to-race ratio is a property of the data, not of the method.

4.3 The gate and the fallback

A curve is issued only if the compound has at least 30 clean long-run laps and its cleaned Friday slope is at least +0.02 s per lap. Otherwise the compound is withheld. Withheld is still a forecast: low degradation, equal to the median race degradation of the other withheld cases this season (leave-one-out), with a 10th to 90th percentile band. Fallback error so far: 0.023 s/lap against 0.106 for the naive line over the 13 withheld cases; the withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13.

4.4 The transfer factor and the bands

For each compound, the ratio of race degradation to cleaned Friday degradation is computed on every completed weekend. For a new weekend the factor is the median of the other weekends' ratios, applied only if at least three exist and a majority sit within ±50% of that median; otherwise the cleaned curve is used as is. Season medians: soft ×0.97, medium ×0.45, hard ×1.17 (hard is not applied because weekends disagree). The 90% band combines the regression's standard error on the slope with resampled ratios from the other weekends.

4.5 Validation protocol

Leave-one-weekend-out: each weekend is predicted from its own Friday and factors learned from the other weekends only, then compared with the race value from the same estimator. We report the mean absolute error for the naive line, the cleaned curve, and Orb v1; the calibration slope and correlation of predicted against observed; bootstrap intervals over compound-weekends; per-compound errors; and a sensitivity sweep over the fuel prior (0.9 to 1.3 kg per lap), the fuel cost (0.025 to 0.035 s per kg), the traffic threshold (20% to 40%) and the minimum run length (4 to 7 laps). Across that sweep the error stays between 0.021 and 0.027 s/lap.

4.6 The fifth confounder: the push profile

The tyre-energy proxy per lap is mass × (the integral of speed squared times path curvature along the lap, plus the integral of the absolute speed change), computed on a uniform 10 m arc-length grid from de-duplicated

position samples so that stale packets do not bias curvature. Two uses. First, the within-run trend of energy against tyre age tells us whether the driver was ramping up: on withheld compounds the median trend is +0.10 MJ per lap of age, on issued compounds 0.00. Second, adding the within-run energy deviation as a covariate gives a push-adjusted slope, degradation at constant push, and its coefficient, the energy price of lap time, has a consistent scale Friday to Sunday, within 0.01 s/MJ at Austria and Barcelona (Austria -0.601 against -0.590 s per MJ, Barcelona -0.578 against -0.575). The push-adjusted slope is a diagnostic and a second opinion, not the headline predictor, because easing off is partly a consequence of degradation and adjusting for it removes some of the signal we want; the as-driven estimator with the learned factor predicts the race better.

4.7 The strategy layer

Compound pace offsets come from each driver's best clean lap per compound in practice or qualifying, whichever the lock records for that weekend (the code tries qualifying first and falls back to practice; in the lock, qualifying supplies the medium offset at Britain, Canada, Miami, Zandvoort, practice or a nominal step supplies the rest), accepted only if the step is between 0.2 and 1.0 s (practice bests on harder compounds are often heavy-fuel laps), otherwise a nominal Pirelli-range step of 0.6 s per compound. With a 21 s pit loss and linear curves, every one-stop and two-stop plan with two or more compounds is enumerated and the fastest is chosen; crossover laps are where the harder compound becomes faster at equal age. Once the race is run, the plan implied by each Friday view is costed under the race-observed curves, so the naive line, the cleaned curve and Orb v1 are compared as decisions, not just as numbers.

4.8 The liquid model

A closed-form continuous-time cell (CfC, Hasani et al. 2022, the fast form of the liquid time-constant network) reads each race stint lap by lap: tyre age, elapsed laps, compound, energy, traffic, temperature, fuel. It predicts the cleaned pace-loss trajectory up to a per-stint constant, exactly the fixed-effect treatment of the linear model, and is scored on held-out stints within each weekend against linear and quadratic fits on the same folds, and against a linear fit given the same per-lap inputs. Result: the per-lap inputs cut the age-only error by about a quarter, lap-weighted, with the three degenerate Hungary cells excluded (31% with them); the network on top of the same inputs beat the best baseline in only 9 of 31 cells (median change in held-out error +2%, no gain). It is kept as the curve-shape tool (accelerating versus settling, cliff lap) and reported honestly.

4.9 The lock file

pipeline.py writes out/lock.json: rules, per-weekend metadata, every validation row, live forecasts, strategy plans. The dashboard and the deck read from it, so every number on screen traces to the lock, a hashed sidecar or a labelled placeholder; the Live Predictor computes its posterior and ranked actions in-process by deterministic replay of the frozen prior and the hashed race file. Re-running the pipeline after a new session (refresh.sh) regenerates everything.

5. Results so far

5.1 Calibration ladder, leave-one-weekend-out

Predictor	Cases	MAE (s/lap)	Calibration slope	Pearson r
Naive pooled fit	29	0.137	+0.07	+0.20
Cleaned Friday curve, issued cases	16	0.045	+0.45	+0.67
Orb v1, issued cases	16	0.023	+0.76	+0.82
Orb v1, all cases incl. low-degradation fallback	29	0.023	+0.77	+0.78

90% bootstrap interval on the Orb v1 error: 0.016 to 0.030 s/lap. Probability that Orb v1 beats the uncalibrated cleaned curve on issued cases: 0.99. Wins over naive: 28 of 29.

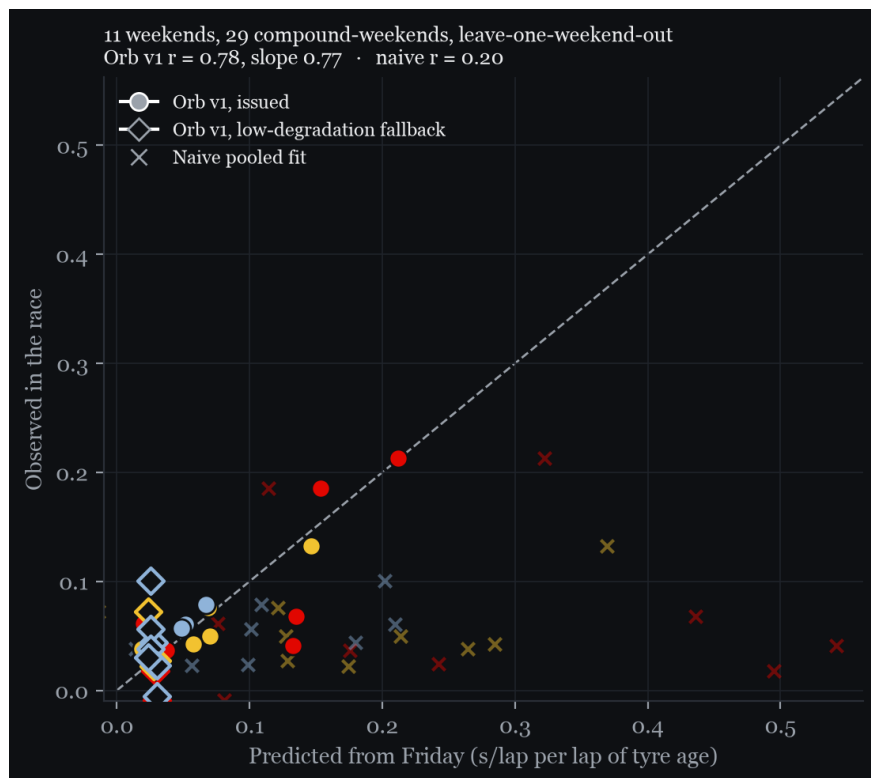


Figure 1. Predicted from Friday against observed in the race, every compound-weekend. Filled circles are issued curves, open diamonds the low-degradation fallback, crosses the naive line. The dashed line is perfect calibration.

5.2 Per compound

Compound	Cases	MAE naive	MAE Orb v1	Transfer factor (median)
Soft	9	0.221	0.032	x0.97
Medium	9	0.148	0.017	x0.45
Hard	11	0.060	0.020	x1.17

5.3 Withheld cases and what the race did

Weekend	Compound	Race degradation (s/lap)
Australia	Medium	+0.022
Australia	Hard	+0.044
Barcelona	Hard	+0.100
Belgium	Hard	+0.056
Britain	Hard	+0.038

Weekend	Compound	Race degradation (s/lap)
Canada	Soft	-0.009
Canada	Hard	-0.005
Japan	Hard	+0.023
Miami	Soft	+0.018
Monza	Medium	+0.028
Monza	Hard	+0.023
Zandvoort	Medium	+0.073
Zandvoort	Hard	+0.030

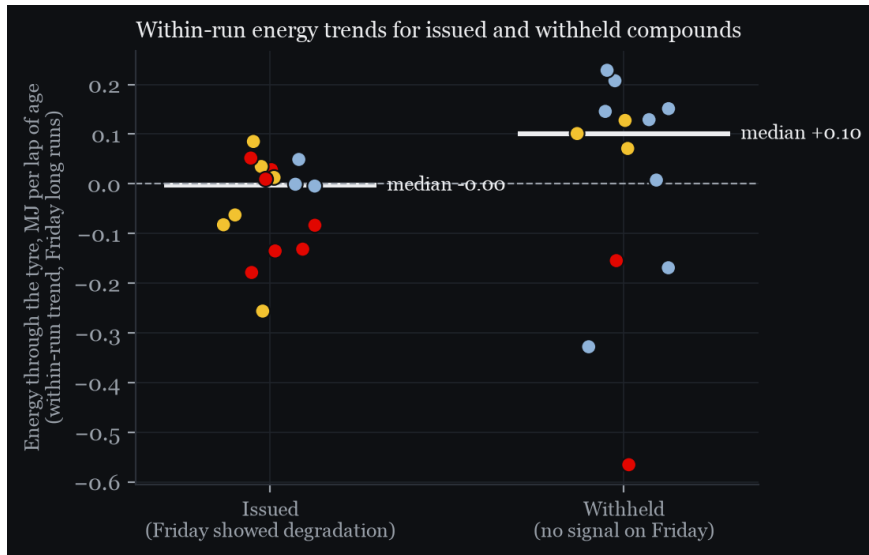


Figure 2. Within-run trend of tyre energy per lap of age on Friday long runs. Withheld compounds are the ones where drivers were still ramping up (median +0.10 MJ per lap); issued compounds are flat (0.00).

5.4 Sensitivity of the headline to the stated choices

Variant	Cases	Issued	MAE Orb v1	r
baseline	29	16	0.023	+0.78
fuel 0.9 kg/lap	29	14	0.026	+0.74
fuel 1.3 kg/lap	29	16	0.023	+0.76
0.025 s/kg	29	14	0.025	+0.75
0.035 s/kg	29	16	0.023	+0.78
traffic ≤20%	29	15	0.027	+0.59
traffic ≤40%	29	16	0.021	+0.79
runs ≥4 laps	31	15	0.026	+0.70
runs ≥7 laps	28	13	0.027	+0.65

5.5 Strategy replay: cost of following each Friday view, under the race-observed curves

Race	Laps	Naive plan	Cost	Orb v1 plan	Cost	Best plan
Australia	58	S-S-M	+14 s	S-M 47/11	+25 s	S-M
Austria	71	M-M-H	+60 s	S-M-M 18/26/27	+26 s	S-M-S
Barcelona	66	S-H-H	+0 s	S-H 12/54	+55 s	S-H-H
Belgium	44	S-S-M	+31 s	M-H 18/26	+10 s	S-M
Canada	68	S-H	+75 s	S-S-H 30/32/6	+30 s	S-H

Race	Laps	Naive plan	Cost	Orb v1 plan	Cost	Best plan
Hungary	70	M-H-H	+74 s	S-M-M 16/26/28	+24 s	S-S-M
Japan	53	M-H-H	+21 s	M-H 43/10	+7 s	M-H
Miami	57	M-H-H	+80 s	S-M 40/17	+6 s	S-M
Monza	53	M-H	+47 s	S-M 44/9	+0 s	S-M
Zandvoort	72	S-M	+155 s	S-S-M 30/32/10	+2 s	S-S-H

Assumptions of the replay: pit loss 21 s, linear curves, no traffic, safety car or weather. Orb v1 beats the naive plan on 8 of the 10 scored weekends; the two misses are Barcelona (+55 s against +0 s for naive) and Australia (+25 s against +14 s for naive); at Barcelona the naive line happened to imply the right plan and Orb v1's soft prediction was low. The replay ranks Friday views as decisions, which is the only ranking a strategist cares about.

5.6 Madrid, live: issued from FP1 and FP2, refreshed after FP3 and after qualifying

Compound	Clean laps	Naive	Cleaned Friday	Orb v1 forecast	90% band	Basis
Soft	46	+0.512	+0.070	+0.068	-0.114 to +0.255	cleaned Friday curve × season transfer factor
Medium	112	+0.153	-0.199	+0.028	-0.013 to +0.088	low-degradation fallback: median race degradation of the 13 withheld cases this season

Medium: energy through the tyre rose +0.48 MJ per lap through the runs, 2.1 times the largest within-run ramp seen at an established circuit this season (+0.23 MJ per lap), which is drivers learning a new track. The push-adjusted second opinion gives +0.033 s/lap, agreeing with the fallback. Hard: no curve, 4 clean laps on Friday (excluded-laps table). Strategy on the central curve: one stop, soft then medium, 22 / 34 laps; the top of the band says two stops. Qualifying refresh landed 21:33 with FP1, FP2, FP3 and Q in the lock; the table shows the post-qualifying lock. Hashed forecast published 21:33 IST (sha256 8e5489d5824a), scored after Sunday's race.

6. What we chose, and the pros and cons of each choice

Choice	Why	Cost or risk	Mitigation
Within-stint fixed-effects regression as the core, not a black-box learner	Identifiable and transparent; identical on practice and race; a Haas engineer can audit every term	Linear in age; no interactions; misses cliffs	Quadratic and liquid shape tools flag the exceptions; linear is kept for prediction because it validates best
Fuel as a stated prior, not estimated from practice	Within a stint fuel and tyre age are collinear; estimating fuel from practice is not identified	A wrong prior biases every slope	Sensitivity sweep 0.9 to 1.3 kg/lap and 0.025 to 0.035 s/kg: error stays 0.023 to 0.026; external race-data estimate 0.0294 s/kg
Track evolution measured from push laps per session	Push laps happen at different times on fresh tyres, so time and age separate	Long-run evolution may differ from push-lap evolution on green tracks	The gate catches it; Madrid medium is the live example
Traffic from the distance-to-car-ahead channel, 30% threshold	Measured per lap, not guessed	Threshold is a choice; NaN gaps drop laps	Sweep 20% to 40%: error 0.027 to 0.021; NaN laps excluded, never assumed clean
A gate that refuses, plus a scored low-degradation fallback	Never silently wrong; the refusal is itself informative	13 of 29 compounds withheld	Fallback error 0.023 s/lap against 0.106 for the naive line over the 13 withheld cases; withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13; push-adjusted second opinion
Transfer factor learned leave-one-weekend-out with an agreement rule	Honest out-of-sample number; applied only where weekends agree	Ten or so weekends is a small sample; hard never qualifies	Bootstrap intervals; 2025 same-track ratios as a prior (downloading)
Tyre-energy proxy from public 3.7 Hz traces	The only public route to the driver's push profile; unique confounder	A proxy: no tyre temperatures or pressures; position feeds sometimes degraded	Arc-length resampling and stale-sample removal; lap-level quality checks; degraded feeds refused
Push-adjusted slope as a diagnostic, not the predictor	Easing off is partly caused by degradation; adjusting removes signal	Two numbers per compound can confuse	Reported as a second opinion with its own factor; the as-driven number is the headline
Public data only	Reproducible for any Grand Prix by name on a laptop; usable by data-poor categories	No fuel flow, tyre temperatures or pressures	Stated priors; the method takes those as inputs when a team has them
Single lock file feeding dashboard and deck	Every number on screen traces to the lock, a hashed sidecar or a labelled placeholder; regeneration is one command	Stale lock if a session lands and the pipeline is not rerun	refresh.sh; the lock carries its timestamp on every page
Streamlit dashboard	Fast to build, interactive, URL-addressable views	Not a production UI; needs the Python environment	Static HTML export as fallback
Liquid network tested and reported as an ablation	Distinctive AI element; honest	No accuracy gain over a linear model with the same inputs	Used for curve shape; the inputs finding is the real result
Solo presenter	One voice, no hand-offs	Bandwidth on the day	Scripted demo path and URL deep links; deck as PDF

6b. Competitive landscape, checked 12 September 09:50 IST

Public GitHub repositories on the TrackShift 2026 tyre problem, plus the one published paper. Finalist status of each team is unknown. Ratings are our reading of their README and artefacts on that morning.

Entry	Method and data	Validation	Practice to race	Honesty and gating	Live forecast	Demo and AI
Orb v1 (us)	Stint fixed-effects on practice and race; fuel prior, measured evolution, traffic, push profile from telemetry energy; 11 scored 2026 weekends, 29 compound-weekends	Leave-one-weekend-out; MAE 0.023 vs 0.137 naive; r 0.78; sensitivity sweep	Learned per compound with an agreement rule; validated	Gate plus validated low-degradation fallback; exclusions listed	Madrid issued from FP1 and FP2, refreshed after FP3 and Q, hashed	Six-view dashboard; liquid-network ablation
PITWALL (OptimistOtaku)	Fixed effects via Frisch-Waugh, cluster-robust errors, fuel estimated from race pit steps, traffic from position; 12 events 2026	Leave-one-event-out on stop value; three published falsifications	Claims no transfer (measured on pit-stop deltas); model omits tyre age	High; states limits	None: replays past races	pptx deck, 8 figures, single-file fallback demo; no learned model
GripTrace (RizaShaik)	Covariate regression; traffic under 200 m as covariate; no fuel model; 2023 Bahrain, soft only, 466 practice laps	Race held out once for the soft	Finds the Friday trend not identifiable	High; careful language	None	Streamlit; no AI element
TyreIQ (suganya1703)	XGBoost on synthetic 2023 Bahrain data with an assumed cliff	In-sample on synthetic data (R-squared 0.98)	None	Low: data is generated	None	Streamlit, four tabs, pit advisor; 9 stars
Dr.Tyre (Shall We Develop)	FastF1 pipeline: fuel, evolution, traffic, ghost baseline; data extent unstated	A validation panel; no numbers in README	None stated	Medium	None	Vite race simulator with an AI race engineer
TIREX (Tanish9022)	Planned Kalman state-space with a physics confounder engine; backend phases 1 and 2 only	Planned	Planned	States data-honesty rules	None	FastAPI backend; no frontend yet
TrackShift AI (Icey067)	Fixed constants for fuel, evolution and dirty air; simulated toggles	In-sample R-squared above 0.94	None	Low	None	React, GSAP, Gemini voice debriefs; Cloud Run demo
Empirical substantiation (vib06hav)	2023 to 2025 practice-only associations	Sensitivity analyses	Concludes not identifiable	High	None	None
arXiv 2512.00640 (Cappello, Hoegh)	Bayesian state-space, one driver, one race, fuel plus latent pace	Against ARIMA on that race	Race only	Academic	None	Code public

Where we lead

- Problem-solution fit: we deliver the literal brief, clean practice curves plus a post-race validation tool, on every compound of every weekend. PITWALL pivoted away from the curve; GripTrace and the substantiation study concluded it cannot be found.
- Scale and protocol of validation: 29 compound-weekends leave-one-weekend-out with a naive baseline, a calibration slope and a sensitivity sweep. In the public repositories we reviewed, we did not find another entry reporting a season-scale held-out score.

- The transfer question answered with data: correlation 0.78 at the compound-weekend level, which is the strategist's unit, against PITWALL's stop-delta measurement whose noise is as large as its signal.
- The fifth confounder, the driver's push profile, measured from telemetry energy: where the cleaned Friday slope pointed the wrong way (9 cases), the driver was ramping up in every one; 4 of the 13 withheld compounds show a falling energy trend and were withheld for too few clean laps; in the public methods and rival repositories we reviewed, we did not find it.
- Refusal as a product feature, with a scored fallback: withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13.
- A live, time-stamped forecast for this weekend; every other entry we reviewed replays a past race or simulates one.

Where they lead, and what we do about it

- PITWALL estimates the fuel cost from race pit steps and uses cluster-robust errors. Answer: our fuel prior is a stated assumption whose sensitivity is published, and it agrees with their estimate (0.030 versus 0.0294 s per kg).
- PITWALL published falsifications of its own strategy ideas. Answer: our Barcelona and Australia misses and the liquid-network ablation are on the slides, not hidden.
- TyreIQ, Dr.Tyre and TrackShift AI have slicker interfaces and race simulators. Answer: our dashboard is plainer but every number on it traces to the lock, a hashed sidecar or a labelled placeholder, and the forecasts are scored; we say that in the first minute.
- TIREX's Kalman design would give principled per-lap uncertainty. Answer: our bands come from regression error plus resampled factors and are checked by coverage; a state-space model is on the post-event roadmap.

Eight questions about the competition, with answers

PITWALL says practice curves do not transfer. Why should we believe you and not them?

Because we measure the transfer at the unit a strategist uses, the compound-weekend slope, and it correlates with the race at 0.78 leave-one-out. Their measurement is on pit-stop deltas, where the standard deviation is nearly the mean, so almost nothing would correlate. Both findings can be true at the same time; ours is the one that answers the brief.

PITWALL estimates fuel from race data and you assume it. Is that not weaker?

It is a stated assumption, and its consequences are published: across 0.9 to 1.3 kg per lap and 0.025 to 0.035 s per kg the error stays between 0.023 and 0.026. Their race-data estimate, 0.0294 s per kg, sits inside our sweep. We chose a prior because within a practice stint fuel and age are collinear and estimation from practice is not identified.

GripTrace found that the tyre-age trend is not identifiable. Did you find the same?

On some weekends, yes: that is exactly what the gate detects, and where the cleaned slope pointed the wrong way we found the reason, the driver ramping up through the run. On the other sixteen compound-weekends the cleaned curve is identifiable and predicts the race. One weekend and one compound cannot tell those two cases apart; twenty-nine can.

TyreIQ reports R-squared 0.98. Yours is far lower. Why?

Theirs is measured on synthetic data generated with an assumed degradation curve, in sample. Ours is measured against eleven real races the model never saw. A high score on your own synthetic data is not evidence about Sunday.

Several rivals have prettier dashboards and race simulators. Why is yours plain?

Because a simulator without a scorecard is a story. Every number on our screens traces to the lock, a hashed sidecar or a labelled placeholder, the forecasts are scored against races, the excluded laps are listed, and the Madrid forecast is on record before the race. We would rather be plain and right than animated and unscored.

What do you have that no rival has?

In the public repositories we reviewed, we did not find any of these four: the fifth confounder measured from telemetry energy, a learned and validated transfer factor, a validated fallback for withheld compounds, and a live time-stamped forecast for this weekend.

What have you borrowed from the rivals?

PITWALL's practice of publishing falsifications, which is why the Barcelona and Australia misses and the network ablation are on our slides; GripTrace's care with language about what public data can and cannot support.

If PITWALL is in the room, what is your one-sentence answer to them?

Friday transfers once you learn how Sunday is managed; here are twenty-nine held-out cases, and here is Madrid on record before the race.

7. Roadmap

See Part A at the front of this document: the full clock-based schedule with task IDs, deliverables, done-criteria, dependencies, fallbacks, the cut list and the rubric map.

8. Twenty questions on technical depth, with answers

1. How do you separate fuel burn from tyre degradation when both change monotonically within a stint?

We do not try to estimate fuel from practice; within a stint the two are collinear and the estimate is not identified. The fuel term is a stated prior from the 2026 regulations, 1.1 kg per lap at 0.030 s per kg, applied as a known offset. The sensitivity sweep shows the headline error stays between 0.023 and 0.026 s/lap across 0.9 to 1.3 kg per lap and 0.025 to 0.035 s per kg, and an independent public estimate from race data gives 0.0294 s per kg.

2. Why a fixed effect per stint rather than per driver?

A stint carries its own fuel load, tyre set, engine mode and driver; a per-driver effect would leave fuel-load differences between runs in the residual and bias the slope. The cost is that we discard the absolute level and keep only the change within a stint, which is exactly what degradation is.

3. How is track evolution identified separately from tyre age?

It is measured per session from every driver's push laps, laps within 1% of that driver's own best, against session time, driver-demeaned. Push laps happen at different times of the session on fresh tyres across drivers, so session time and tyre age are not collinear. At Madrid it was 0.6 to 0.8 s per hour on Friday (FP1 0.63, FP2 0.76 s per hour).

4. How do you detect traffic, and how sensitive is the result to that choice?

From the distance-to-car-ahead channel at 3.7 Hz: the share of the lap spent within 60 m of the car ahead. Laps above 30% are dropped; laps without gap data are dropped rather than assumed clean. At 20% the error is 0.027, at 40% it is 0.021.

5. Why leave-one-weekend-out rather than ordinary cross-validation?

Because the deployment question is a new weekend with no race yet. A random split would let a weekend's own race leak into the factor applied to its Friday. Every number in the deck is computed with the held-out weekend's race unseen.

6. Your transfer factor is a median over ten numbers. Is that not fragile?

It is small, which is why it is applied only under an agreement rule, at least three weekends with a majority within 50% of the median, why hard never qualifies, and why we show bootstrap intervals: 0.016 to 0.030 on the error. The 2025 same-track ratios are the next test and are downloading.

7. What does withheld mean, and is it not a way to avoid being wrong?

It means Friday had fewer than 30 clean long-run laps or no positive cleaned slope. It is still a forecast: low degradation, the median race value of the other withheld cases, leave-one-out. That forecast has an error of 0.023 s/lap against 0.106 for the naive line over the 13 withheld cases; the withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13. It is scored like everything else.

8. Why does a cleaned practice curve come out negative on some compounds?

Because the driver was ramping up. The within-run trend of tyre energy per lap of age is +0.10 MJ per lap at the median on withheld compounds and 0.00 on issued ones. Lap times fall faster than fuel burn explains, and the estimator reads it as negative degradation. We measure that rather than assume it.

9. What is the energy proxy and how do you trust 3.7 Hz position data?

Mass times the integral of speed squared times path curvature along the lap, plus the integral of absolute speed change: lateral and longitudinal work through the contact patch. Curvature is computed on a 10 m arc-length grid from de-duplicated samples. Each lap carries a feed-quality check; the Hungary race feed had about 26 distinct points per lap instead of 300, so its energy is pace-estimated and its traffic is unknown, and the pipeline says so.

10. Is the energy covariate not endogenous? Drivers lift because the tyre is degrading.

Yes, partly, which is why the push-adjusted slope is a diagnostic and a second opinion and not the predictor. Adjusting for push removes some of the degradation signal we want, and the as-driven estimator with the learned factor predicts the race better: the probability that the push-adjusted version beats it is 0.14.

11. Where is the machine learning?

The transfer factor and the fallback are parameters learned on training weekends and scored on held-out weekends. The liquid network was trained and tested with a fair ablation. The AI that earns its place is the part that survives the held-out test.

12. Tell us honestly what the liquid network did.

A closed-form continuous-time cell read each race stint lap by lap and predicted the cleaned trajectory, scored on held-out stints. Against age-only fits it wins almost everywhere. Against a linear model given the same per-lap inputs it beats the best baseline in only 9 of 31 cells, and its median gain is about zero (+2% change in held-out error). The inputs were the discovery: energy, traffic and fuel cut the age-only error by about a quarter, lap-weighted, with the three degenerate Hungary cells excluded (31% with them). The network stays as the curve-shape tool.

13. How do you validate the uncertainty bands?

By coverage: the share of 90% bands that contain the race value, computed leave-one-out. With 16 issued cases the check is coarse, so it is reported as a count, not a percentage; the band construction combines the regression's standard error with resampled transfer ratios from other weekends.

14. How does this fail?

New circuits and green tracks, where drivers ramp up: the gate withholds and the fallback answers, and Madrid is the live example. Wet sessions: excluded. Degraded telemetry feeds: refused, not modelled. Safety cars and traffic in the race: the strategy replay ignores them and says so. Compounds with thin running, hard at most European rounds: withheld for too few laps.

15. Why linear degradation curves?

Because linear validates best for prediction at the tyre ages actually run. The quadratic and liquid curves split the season 15 accelerating against 13 settling, so there is no single non-linear shape to assume. The shape tools flag the exceptions and the cliff lap where one exists.

16. How does this compare with the published state-space model on public F1 data?

That model (arXiv 2512.00640) fits one driver's race with fuel and a latent pace, with no practice input, no traffic, no track evolution and no cross-weekend validation. Ours starts from practice, strips four measured confounders and one diagnostic one, and is validated leave-one-weekend-out across eleven weekends. Its code is public and we can run it as a baseline on one race.

17. A public analysis concludes practice curves do not transfer to the race. Your response?

That conclusion is measured on stop-by-stop pit deltas, a target whose standard deviation is close to its mean. At the compound-weekend level, with the management factor learned leave-one-out, the correlation between Friday and Sunday is 0.78 against 0.20 for the naive line. The curve transfers once you learn how each compound is managed.

18. What does the strategist actually get on Saturday night?

Per compound: the slope, the 90% band, the basis in one sentence, or the withheld reason and the low-degradation forecast. Crossover laps. The best one-stop and two-stop plans with stint lengths and what flips the call. The list of excluded laps with reasons.

19. How and when is the Madrid forecast scored?

The forecast was issued from FP1 and FP2 and is time-stamped; qualifying refresh landed 21:33 with FP1, FP2, FP3 and Q in the lock; hashed forecast published 21:33 IST (sha256 8e5489d5824a). After Sunday's race, which starts 18:30 IST, after the event closes, the same estimator runs on the race laps and the scorecard is published. We commit to publishing it whether it is right or wrong.

20. What would you need from a team to make this production-grade?

Fuel mass per lap, tyre pressures and temperatures, and the driver's push instructions. The fuel prior and the energy proxy then become measured inputs, the gate fires less often, and the rest of the method is unchanged. For F2, F3 and F1 Academy, which have public timing and two engineers, it runs as is.

Appendix

A. Files and commands

File	Role
features.py	Per-lap features from telemetry (pre-work)
model_v2.py	Stint fixed-effects estimator, practice and race (pre-work)
extract_all.py, extract_more.py, extract_extra.py	Download and feature extraction per weekend and year
pipeline.py	Every weekend -> out/lock.json, results.csv, validation.csv, excluded_<event>.csv
strategy2.py	Offsets, plans, crossover, replay scoring
liquid.py	CfC shape model with held-out ablation -> out/liquid.json
app.py	Streamlit dashboard (streamlit run app.py; views at ?ev=<event>&view=<view>)
refresh.sh	Pull new Madrid sessions and rebuild the lock
build_deck*.py, talk_track.md	Deck and script

B. Stated assumptions

- Fuel: 1.1 kg per lap in practice at 0.030 s per kg; race fuel 70 kg burned linearly over the race distance.
- Traffic: laps with more than 30% of the distance within 60 m of the car ahead are dropped.
- Runs: at least 5 clean laps; laps slower than 105% of the run best dropped.
- Gate: at least 30 clean laps and a cleaned slope of at least +0.02 s/lap.
- Transfer factor: median of other weekends, applied only with at least three and a majority within 50%.
- Strategy: pit loss 21 s; compound offsets from practice or qualifying medians if the step is 0.2 to 1.0 s, else a nominal Pirelli-range step of 0.6 s per compound; linear curves; no safety car, traffic or weather.
- Madrid race length 56 laps (FIA rule, 5.474 km circuit).

C. Data sources

- FastF1 3.8.3: 2018 to today, laps, telemetry, position, weather, race control. Used for all 2026 rounds that exist (Bahrain and Saudi Arabia do not) and fourteen 2025 rounds.
- OpenF1: intervals, pit, stints, race control, 2023 onward.
- Pirelli press releases (compound nominations: Madrid C2 hard, C3 medium, C4 soft). FIA event documents (tyre allocation, pit stop summaries).
- TUMFTM racetrack-database and race-simulation (track centrelines, published tyre and fuel parameters). Published baseline code: colecappello12/F1_SSM_Paper.

D. Known limits

- Hard compound is often withheld for too few Friday laps at European rounds.
- The Hungary race and China FP1 position feeds are degraded at source and are refused for telemetry quantities.
- Sprint weekends have one practice session; the sprint itself is not yet used as a validation target.
- Bands are wide where a compound has few runs (Madrid soft: 46 laps).
- The replay ignores safety cars, traffic and weather, and uses linear curves.